
Online Purchasing Behavior

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1. Problem Description

- In today's digital world, e-commerce platforms generate a vast amount of customer interaction data such as browsing time, clicks, and product views. However, predicting whether a customer will make a purchase remains a challenge. This project aims to build a machine learning model that classifies user behavior into purchasing or non-purchasing categories based on their online activity.
- The system analyzes features like session duration, number of clicks, pages visited, and interactions. By applying data preprocessing techniques such as scaling and feature selection, irrelevant or low-impact data is removed to improve performance.
- This project has real-life applications in improving targeted marketing, recommendation systems, and increasing conversion rates for online businesses. It helps companies understand customer intent and optimize their sales strategies effectively.

1.1 Dataset Description

- The dataset used in this project contains information about user browsing behavior on an e-commerce website. It is designed to help predict whether a user will make a purchase (generate revenue) or not.

1.2 Flow Diagram



Fig.1 Sample flow diagram.

2. Modules for Project Implementation

2.1 Modules Used(mention all the inbuilt/ custom modules used and the purpose)

The following Python modules were used in the implementation of the project:

1. Pandas (pandas)

- **Purpose:** Data handling and manipulation
- **Usage in project:**
 - Reading CSV file (read_csv)
 - Handling missing values
 - Removing duplicates
 - Data filtering and transformation

2. Matplotlib (matplotlib.pyplot)

- **Purpose:** Data visualization
- **Usage in project:**
 - Plotting bar graph (class distribution)
 - Plotting accuracy comparison graph

- Plotting ROC curve

3. Scikit-learn (sklearn)

a) Preprocessing Module

- **LabelEncoder**
 - Converts categorical data into numeric form
- **StandardScaler**
 - Scales features for better model performance

b) Model Selection Module

- **Train_test_split**
 - Splits dataset into training and testing sets (70–30)

c) Machine Learning Models

- **LogisticRegression**
 - Used for binary classification
 - Predicts whether a user will purchase or not
- **KNeighborsClassifier (KNN)**
 - Classifies data based on nearest neighbors by distance

d) Evaluation Metrics

- **Accuracy_score**
 - Measures overall correctness of the model
- **Classification_report**
 - Provides precision, recall, and F1-score
- **Confusion_matrix**
 - Shows correct and incorrect predictions
- **ConfusionMatrixDisplay**
 - Used to visualize confusion matrix

e) ROC Curve Functions

- **ROC curve**
 - Calculates True Positive Rate and False Positive Rate
- **AUC score**
 - Computes Area Under Curve (AUC score)

2.2 Platform Used: (mention all the platforms with the reason of using it)

- **Kaggle:** for searching the data set
- **Jupyter:** For code writing, testing, and debugging.

3. Machine Learning Model/ Searching Technique Used(Describe

each ML model/ searching technique used)

In this project, two machine learning models were used to predict whether a user will make a purchase or not: Logistic Regression and K-Nearest Neighbors (KNN).

3.1 Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for binary classification problems. It predicts the probability of a data point belonging to a particular class.

In this project, Logistic Regression is used to classify whether a user will generate revenue (1) or not (0).

The model uses the sigmoid function to convert output values into probabilities between 0 and 1. Based on a threshold (usually 0.5), the final class label is assigned.

Key Features:

- Suitable for binary classification
- Fast and efficient
- Provides probability output
- Works well with linearly separable data

Role in Project:

- Trained on preprocessed and scaled data
- Used to predict purchase behavior
- Achieved higher accuracy compared to KNN

3.2 K-Nearest Neighbors (KNN)

K-Nearest Neighbors (KNN) is a supervised machine learning algorithm used for classification and regression. It works based on the concept of similarity.

It classifies a data point by looking at the K nearest data points (neighbors) and assigning the most common class among them.

In this project, KNN uses Euclidean distance to find the nearest neighbors and predict whether a user will make a purchase.

Key Features:

- Simple and easy to understand
- No training phase (lazy learning)
- Works well with small datasets
- Sensitive to feature scaling

Role in Project:

- Used as a comparison model
- Applied after feature scaling
- Accuracy was lower than Logistic Regression

3.3 Model Comparison

Both models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

The results showed that Logistic Regression performed better than KNN for this dataset, making it the preferred model.

4. Evaluation metrics used (Describe each evaluation metric used along with formulae)

To evaluate the performance of the machine learning models, several evaluation metrics were used. These metrics help measure how accurately the model predicts whether a user will make a purchase or not.

4.1 Accuracy

Accuracy measures the overall correctness of the model by calculating the ratio of correctly predicted observations to the total observations.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Where:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

It tells how many predictions were correct out of all predictions.

4.2 Precision

Precision measures how many of the predicted positive cases are actually positive.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Explanation:

It focuses on the accuracy of positive predictions.

4.3 Recall (Sensitivity)

Recall measures how many actual positive cases are correctly identified by the model.

$$\text{Recall} = \frac{TP}{TP + FN}$$

Explanation:

It shows how well the model detects actual positive cases.

4.4 F1-Score

F1-Score is the harmonic mean of Precision and Recall. It provides a balance between both metrics.

$$F1 = 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$$

Explanation:

Used when there is class imbalance and both precision and recall are important.

4.5 Confusion Matrix

A confusion matrix is a table used to evaluate the performance of a classification model.

	Predicted Positive	Predicted Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

Explanation:

It provides a detailed breakdown of correct and incorrect predictions.

4.6 ROC Curve (Receiver Operating Characteristic Curve)

ROC curve is a graphical representation that shows the performance of a classification model at different thresholds.

$$TPR = \frac{TP}{TP + FN}, FPR = \frac{FP}{FP + TN}$$

Explanation:

- X-axis → False Positive Rate (FPR)

- Y-axis $\rightarrow$ True Positive Rate (TPR)
- A curve closer to the top-left corner indicates better performance

4.7 AUC (Area Under Curve)

AUC represents the area under the ROC curve.

Explanation:

- $AUC = 1 \rightarrow$ Perfect model
- $AUC = 0.5 \rightarrow$ Random model
- Higher AUC $\rightarrow$ Better model performance

5. Results and Visualizations

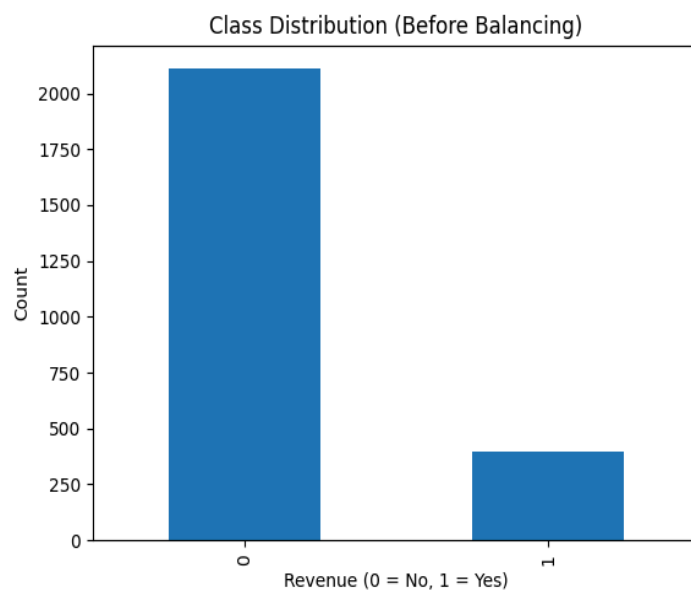


Fig.1 class Distribution (before balancing)

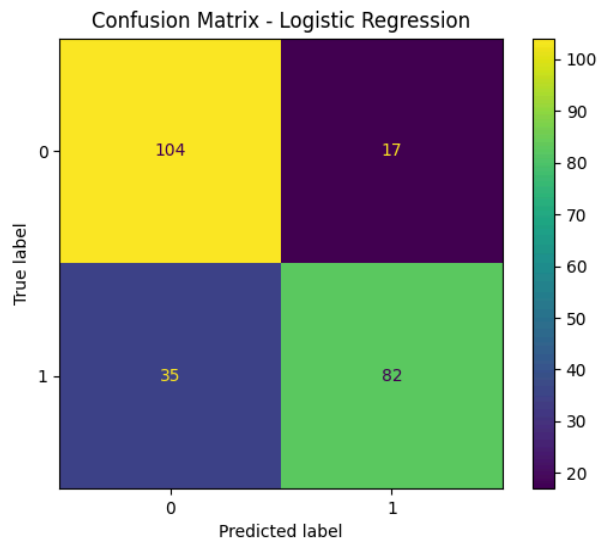


Fig.2 Confusion Matrix Graph

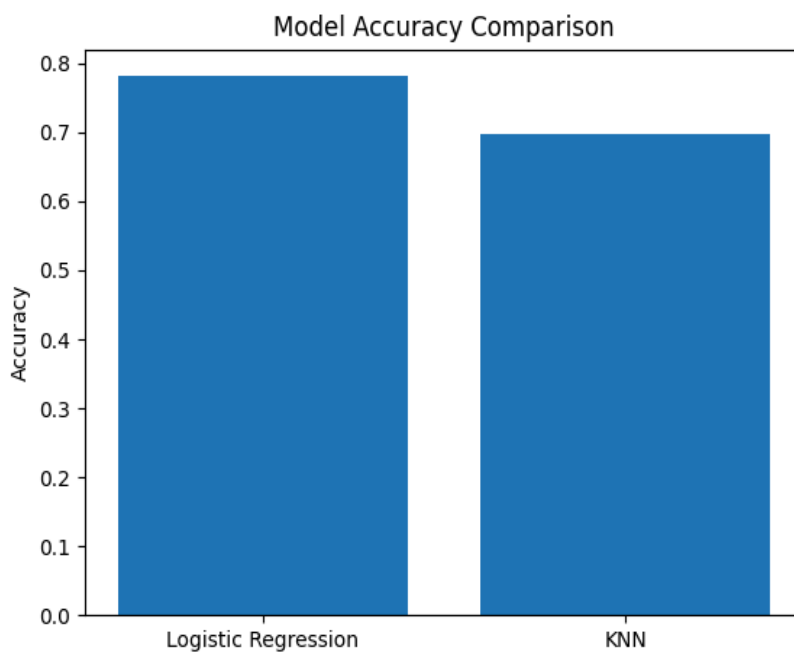


Fig. 3 Model Accuracy Comparison Graph

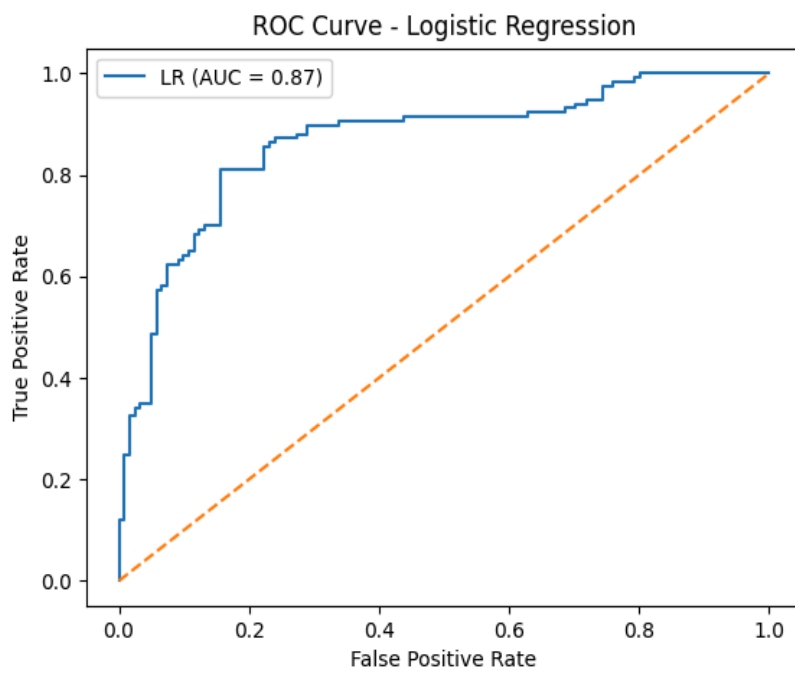


Fig.4 ROC(Receiver Operating Characteristic) Curve Graph

6. Conclusion and Future Scope:

Conclusion

- This project successfully developed a machine learning model to predict whether a user will make a purchase based on their browsing behavior on an e-commerce platform.
- Data preprocessing techniques such as handling missing values, removing duplicates, encoding categorical variables, and feature scaling were applied to improve data quality. The dataset was also balanced using undersampling to address class imbalance.
- Two machine learning models, Logistic Regression and K-Nearest Neighbors (KNN), were implemented and evaluated using metrics such as accuracy, precision, recall, F1-score, confusion matrix, and ROC curve.
- The results showed that Logistic Regression performed better than KNN, achieving higher accuracy and more reliable predictions. Therefore, Logistic Regression was selected as the final model for this project.
- Overall, the system demonstrates how machine learning can be effectively used to analyze user behavior and support decision-making in e-commerce platforms.

Future Scope

Although the current model performs well, there are several ways to further improve and extend this project:

- Use advanced models such as Decision Tree, Random Forest, or Support Vector Machine (SVM) to improve accuracy
- Apply hyperparameter tuning techniques to optimize model performance
- Use larger and more diverse datasets for better generalization
- Implement real-time prediction system for live user data
- Develop a user interface or web application for easier interaction
- Explore deep learning techniques for more complex pattern recognition

7. References:

Books:

1. Let Us Python – Yashavant Kanetkar

Online Resources:

- GeeksforGeeks -- <https://www.geeksforgeeks.org/>
- TutorialsPoint -- <https://www.tutorialspoint.com/cprogramming/index.htm>
- Kaggle -- <https://www.kaggle.com/>